UNIT 4 · TOPIC 3(A)

Power, Torque and Speed Calculations

Syllabus topic: Calculations of power, torque, speed; firing arrangement and intervals

Core formulas

$$\text{BP (kW)} = 2\pi NT/60{,}000 \Leftrightarrow \text{T (N·m)} = 9550 \times \text{P (kW)}/\text{N (rpm)}$$

$$\text{IP (kW)} = p_m \cdot L \cdot A \cdot n \cdot K/60{,}000$$

p_m in kPa; n = power strokes/min = N/2 for a 4-stroke engine; K = number of cylinders.

$$\text{FP} = \text{IP} - \text{BP};\ \eta_{mech} = \text{BP}/\text{IP};\ 1\ \text{hp} = 0.746\ \text{kW}$$

$$\text{Mean piston speed} = 2LN/60\ \text{(m/s)}$$

A wear and inertia criterion — tractor diesels run at ~**6–12 m/s**.

$$\text{Torque backup (torque rise)} = (T_{max} - T_{rated})/T_{rated} \times 100$$

Higher backup (**15–30%**) means better LUGGING: as the load slows the engine, torque RISES and the engine "hangs on" instead of stalling.

Worked examples

Ex.1 BP = 45 kW at 2200 rpm → T = 9550 × 45/2200 = **195 N·m**.

Ex.2 (IP from MEP) p_m = 700 kPa, D = 100 mm (A = 78.54 cm²), L = 120 mm, 4 cylinders, N = 2200 rpm.
1. Work per power stroke per cylinder = 700 × 0.12 × 0.007854 = **0.66 kJ**.
2. n = 2200/2 = 1100 power strokes/min.
3. IP per cylinder = 0.66 × 1100/60 = **12.1 kW** → engine IP = **48.4 kW**.
4. If BP = 40 kW → η_mech = 40/48.4 = **82.6%**.

Ex.3 Mean piston speed = 2 × 0.12 × 2200/60 = **8.8 m/s**.

Ex.4 (torque backup) T_rated = 195 N·m at 2200 rpm, T_max = 240 N·m at 1400 rpm → backup = (240 - 195)/195 = **23%**.

Practice MCQs

Q71. Torque (N·m) from power P (kW) at N rpm is:

(a) 9550·P/N (b) P·N/9550
(c) 60·P/N (d) 746·P/N

Q72. BP = 45 kW at 2200 rpm gives torque ≈

(a) 145 (b) 175
(c) 195 (d) 225 N·m

Q73. In the IP formula, n for a 4-stroke engine equals:

(a) N (b) N/2
(c) 2N (d) N/4

Q74. p_m = 700 kPa, L = 0.12 m, A = 78.54 cm², 4 cylinders, 2200 rpm → IP ≈

(a) 24 kW (b) 36 kW
(c) 48 kW (d) 60 kW

Q75. If BP = 40 kW for the engine of Q4, mechanical efficiency ≈

(a) 72% (b) 78%
(c) 83% (d) 90%

Q76. Mean piston speed equals:

(a) LN/60 (b) 2LN/60
(c) πLN/60 (d) 4LN/60

Q77. L = 120 mm at 2200 rpm → mean piston speed =

(a) 4.4 (b) 8.8
(c) 13.2 (d) 17.6 m/s

Q78. Torque backup of an engine indicates its:

(a) Fuel economy (b) Lugging ability under overload
(c) Oil pressure (d) Idle quality

Q79. T_{rated} = 195 N·m, T_{max} = 240 N·m → torque backup ≈

(a) 12% (b) 18%
(c) 23% (d) 45%

Q80. As load slows a diesel below rated speed, its torque:

(a) Falls (b) Rises toward T_{max}
(c) Is constant (d) Becomes zero

Answers

Q71 — (a) 9550·P/N
Q72 — (c) 195 — *T = 9550 × 45/2200 = 195.3 N·m.*
Q73 — (b) N/2
Q74 — (c) 48 kW — *0.66 kJ/stroke × 1100/60 × 4 = 48.4 kW.*
Q75 — (c) 83% — *40/48.4 = 82.6%.*
Q76 — (b) 2LN/60
Q77 — (b) 8.8 — *2 × 0.12 × 2200/60 = 8.8 m/s.*
Q78 — (b) Lugging ability under overload
Q79 — (c) 23% — *45/195 = 23%.*
Q80 — (b) Rises toward T_{max}

Sources: Liljedahl et al. — Tractors and their Power Units; standard IC-engine texts; Kepner et al.; NPTEL farm power lectures.